👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.8

Solve and Graph Inequalities

Lesson 8-6 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve an inequality and graph its solution set on a number line.

Language Objective: I can explain my work using the words solve, graph, solution, and substitute.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Solve · **Graph** · **Solution** · **Substitute** · **Solution set** · **Inverse operation**

Level 3: try the blanks from memory first, then check the bank.

1. To find all the values that make an inequality true is to _____ it.

2. To show the solutions on a number line is to _____ them.

3. A value that makes an inequality true is a _____ .

4. To check a solution, I _____ the value back into the inequality.

5. All the values that make an inequality true form the _____.

6. An operation that undoes another to isolate the variable is an _____.

Write About the Math

How much more does the detective need? How did you solve and graph this inequality?
¿Cuánto más necesita la detective? ¿Cómo resolviste y graficaste esta desigualdad?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Solve** — *Resolver*
- ☐ **Graph** — *Graficar*
- ☐ **Solution** — *Solución*
- ☐ **Substitute** — *Sustituir*
- ☐ **Solution set** — *Conjunto solución*

Model: *Solving an inequality gives a range of solutions, not one number.* — Students note both use inverse operations to isolate the variable, but an inequality's solution is many values (a range) while an equation's is a single value.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective needs at least $__ more because 48 + s ≥ 75 means s ≥ 75 − ___ = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Solve
2. **Notes 2:** Graph
3. **Notes 3:** Solution
4. **Notes 4:** Substitute
5. **Notes 5:** Solution set
6. **Notes 6:** Inverse operation

7. **Watch for this mistake:** *A common mistake in Solve and Graph Inequalities is shading the number line in the wrong direction after solving. For example, a student solves x + 4 > 10 correctly to get x > 6, but then shades LEFT toward smaller numbers instead of RIGHT toward greater numbers. The shading direction always matches the solved symbol, not the original problem: '>' and '≥' shade right (greater numbers), '<' and '≤' shade left (smaller numbers) — check this AFTER you solve, not before. To catch this mistake, substitute one shaded value and one unshaded value back into the original inequality: the shaded value should make it true, and the unshaded value should make it false.*
8. **Try It 1:** A. Yes, because 5 ≤ 5 is true — *Substitute x = 5 into x ≤ 5: 5 ≤ 5 is true because ≤ means 'less than OR equal to.' The boundary value 5 IS a solution, so the gate opens.*
9. **Try It 2:** A. All numbers greater than 7 — *The solution set is every value that makes x > 7 true. That means all numbers greater than 7, such as 8, 9, 10, and beyond. Because > does not include the boundary, 7 itself is NOT in the solution set.*